

1. For each of the given rings R and subsets S , prove that S is a subring of R .
 - (a) Let $R = M_2(\mathbb{R})$ and $S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R}, a + c = b + d \right\}$.
 - (b) Let R be an arbitrary ring and $a \in R$, define $S = \{r \in R : ar = 0\} \cup \{1\}$.
 - (c) Let $R = \mathbb{Z} \times \mathbb{Z}$ and $S = \{(x, y) \in R : x - y = 0\}$.
 - (d) Let $R = \mathbb{R}$ and $S = \{x + \sqrt{3}y : x, y \in \mathbb{Z}\}$.
 2. Let R be a ring with subrings S_1 and S_2 . Prove or disprove the following:
 - (a) $S_1 \cup S_2$ is a subring of R .
 - (b) $S_1 \cap S_2$ is a subring of R .
 - (c) $S_1 \times S_2$ is a subring of $R \times R$.
 3. Let R and S be rings.
 - (a) Prove that $R \times S$ is a ring.
 - (b) Prove that $R \times S$ is commutative if and only if R and S are both commutative.
 4. Let R be a ring and $r \in R$. Prove that if r is a unit, then r cannot be a zero divisor.
 5. Determine whether the following statements are true or false:
 - (a) The trivial ring is commutative.
 - (b) The trivial ring is a subring of every ring.
 - (c) The subring of a domain is a domain.
 - (d) The subring of a division ring is a division ring.
- NOTE: The trivial ring is $\{0\}$ where $0 + 0 = 0$ and $0 \cdot 0 = 0$. The element 0 is both the zero and the identity.
6. Let R and S be rings.
 - (a) Suppose $R \times S$ is an integral domain.

-
- (i) Prove that at least one of R and S is an integral domain.

HINT: Proof by contradiction.

- (ii) Assume, without loss of generality, that R is an integral domain. Prove that $S = \{0\}$.

- (b) Prove that if R is an integral domain and $S = \{0\}$ then $R \times S$ is an integral domain.

- (c) Find a counterexample to the following statement: If R and S are fields, then $R \times S$ is a field.

7. A ring R is called **Boolean** if $x^2 = x$ for all $x \in R$.

- (a) Prove that, in a Boolean ring, $r + r = 0$ for all $r \in R$.

HINT: Take $x = r + r$ and apply the identity $x^2 = x$.

- (b) Show that, in a Boolean ring, $r = -r$ for all $r \in R$.

- (c) Prove that every Boolean ring is commutative.

HINT: Take $x = r + s$ and apply the identity $x^2 = x$, use the previous results too.

8. Let R be a ring. Suppose that $x^2 - x \in Z(R)$ for all $x \in R$.

- (a) Prove that $sr + rs \in Z(R)$ for all $r, s \in R$.

HINT: Take $x = r + s$ and use the fact that $x^2 - x \in Z(R)$.

- (b) Prove that $r^2 \in Z(R)$ for all $r \in R$ and conclude that R is commutative.

HINT: $Z(R)$ is a subring of R , so it is closed under addition.